

Topic and Problem Statement

Project Title

SmartNShine ATS Resume Generator with AI Interview Practice

Date

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Topic

AI-Powered Career Preparation Platform: ATS Resume Generation and Voice-Enabled Mock Interview Practice Using Large Language Models

Problem Statement

The modern job application process has become increasingly difficult to navigate for the average job seeker. Two specific barriers stand out as the most common reasons qualified candidates fail to secure opportunities even before reaching a human interviewer.

The first barrier is Applicant Tracking Systems (ATS). Most large organizations use ATS software to filter incoming resumes automatically before any human review takes place. These systems parse resume content and score it against job description keywords, formatting rules, and structural expectations. A resume that is poorly formatted, uses non-standard section headings, or lacks the right keywords is rejected outright — regardless of the candidate's actual qualifications. Studies estimate that over 75% of resumes are discarded by ATS filters before a recruiter ever sees them. Despite this, the majority of job seekers have no reliable tool to measure or improve their resume's ATS compatibility before applying.

The second barrier is interview unpreparedness. Even candidates who pass ATS screening frequently struggle in interviews due to lack of structured practice. Generic interview preparation resources provide standard questions but cannot personalize feedback based on the candidate's actual experience, target role, or specific weaknesses. Interview coaching is expensive and inaccessible to most students and early-career professionals. As a result, candidates enter interviews without knowing how they are likely to perform or what specific areas need improvement.

Existing tools address these problems in isolation. Resume builders help with formatting but provide no ATS feedback. ATS checkers score resumes but cannot improve them. Interview platforms offer practice questions but do not evaluate voice responses, generate role-specific questions, or connect the interview context to the candidate's own resume. No widely accessible platform integrates resume optimization and interview preparation into a single end-to-end workflow.

SmartNShine addresses this gap by providing a unified platform where a job seeker can build and optimize an ATS-friendly resume using AI, receive a detailed ATS compatibility score against a specific job description, and then directly practice a mock interview — driven by questions generated from their own resume and target role — with voice input support, multi-dimensional answer evaluation, and a personalized

performance report. The platform removes the cost barrier through a freemium subscription model, making AI-assisted career preparation accessible to students, fresh graduates, and working professionals alike.

Objectives Derived from the Problem

1. To enable users to create structured, ATS-compatible resumes without requiring formatting expertise.
2. To use AI to enhance resume language, categorize skills, and generate professional summaries automatically.
3. To analyze the alignment between a user's resume and a target job description, providing a measurable ATS score and actionable keyword feedback.
4. To generate personalized interview questions based on the user's resume content, job role, seniority level, and interview type.
5. To evaluate user answers — both typed and voice-based — across multiple performance dimensions and provide specific improvement guidance.
6. To deliver a comprehensive post-interview report that helps the user understand their strengths and target their preparation.
7. To make these capabilities accessible through a subscription model that allows free usage within limits and unlocks advanced features for premium users.